

VITIM, Russian Federation

WMO: 300540

Lat: 59.4605N

Lon: 112.5670E

Elev: 190

StdP: 99.06

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -47.4	(c) -45.0	(d) -49.5	(e) 0.0	(f) -47.5	(g) -47.1	(h) 0.0	(i) -44.9	(j) 7.8	(k) -16.5	(l) 7.0	(m) -18.3	(n) 0.4	(o) 340	(p) 0.702

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	29.9	19.3	27.7	18.5	25.6	17.6	20.6	27.1	19.5	25.6	18.4	24.1	2.7	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.6	23.5	17.2	12.6	22.2	16.2	11.8	21.2	60.6	27.1	56.7	25.7	53.2	24.2	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.5	6.4	5.9	-49.1	33.7	3.2	1.7	-51.4	34.9	-53.3	35.9	-55.0	36.9	-57.4	38.1		
			-49.3	22.4	3.2	1.3	-51.6	23.4	-53.5	24.2	-55.3	24.9	-57.6	25.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.0	-28.3	-22.7	-12.8	-1.6	6.9	16.0	18.7	15.5	7.2	-3.0	-17.2	-27.5
	DBStd	18.01	10.38	9.01	7.99	4.98	4.15	4.15	3.27	3.50	4.10	5.73	10.06	10.67
	HDD10.0	5763	1188	915	706	347	114	4	0	3	104	402	817	1162
	HDD18.3	8220	1446	1148	964	597	354	92	36	101	336	661	1067	1420
	CDD10.0	665	0	0	0	0	18	185	269	173	19	0	0	0
	CDD18.3	81	0	0	0	0	0	22	47	12	0	0	0	0
	CDH23.3	987	0	0	0	0	14	340	493	136	3	0	0	0
	CDH26.7	284	0	0	0	0	2	105	149	28	0	0	0	0
(14) Wind	WSAvg	2.4	1.9	2.2	2.3	2.7	2.8	2.5	2.4	2.3	2.6	2.7	2.4	1.9
(15) Precipitation	PrecAvg	439	27	19	15	18	36	51	59	59	48	40	39	33
	PrecMax	648	60	38	37	44	90	123	144	160	106	100	69	59
	PrecMin	267	7	2	3	1	8	1	0	9	12	10	18	10
	PrecStd	76	11	9	8	8	15	27	35	29	22	19	12	11
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.0	-1.2	7.5	14.4	25.2	32.3	33.1	30.0	21.7	10.8	3.1	-2.3
		MCWB	-6.2	-2.7	2.5	6.9	14.8	19.5	20.9	20.7	14.0	6.3	1.4	-3.3
	2%	DB	-7.6	-3.7	3.2	11.2	21.0	29.8	30.3	26.7	18.2	7.9	1.2	-5.9
		MCWB	-8.4	-4.9	-0.1	5.0	11.7	18.6	19.8	18.5	11.8	4.8	0.1	-6.7
	5%	DB	-10.2	-6.8	0.7	8.5	18.5	27.4	28.2	24.6	15.6	5.7	-0.3	-8.6
		MCWB	-10.8	-7.7	-1.6	3.4	10.1	17.6	19.0	17.4	10.8	3.4	-1.4	-9.3
	10%	DB	-13.2	-9.8	-1.4	6.1	15.7	25.0	26.1	22.3	13.4	4.0	-2.7	-11.8
		MCWB	-13.7	-10.5	-3.4	1.8	9.0	16.6	18.4	16.2	10.0	2.2	-3.7	-12.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.9	-2.6	2.5	7.4	15.4	21.6	22.2	21.4	14.8	6.9	1.8	-3.4
		MCDB	-5.1	-1.5	6.8	13.6	23.5	29.7	30.2	27.6	20.3	9.4	3.1	-2.6
	2%	WB	-8.3	-4.8	0.3	5.4	12.9	19.7	21.1	19.5	13.0	5.3	0.3	-6.6
		MCDB	-7.6	-3.7	3.0	10.6	19.2	26.9	27.6	25.0	16.4	7.5	1.1	-6.0
	5%	WB	-10.8	-7.8	-1.6	3.7	11.0	18.4	20.0	18.3	11.6	3.7	-1.4	-9.3
		MCDB	-10.1	-6.9	0.6	7.8	17.4	25.5	26.2	23.0	14.8	5.3	-0.3	-8.6
	10%	WB	-13.8	-10.5	-3.4	2.3	9.3	17.2	19.1	17.1	10.2	2.3	-3.7	-12.3
		MCDB	-13.2	-9.7	-1.5	5.5	15.2	23.7	24.8	21.1	13.1	3.8	-2.7	-11.7
(35) Mean Daily Temperature Range	MDBR		8.5	10.8	14.4	11.6	11.9	12.5	11.3	10.6	8.2	6.6	7.7	7.7
	5% DB	MCDBR	9.9	9.6	12.8	14.7	18.1	17.8	16.3	15.5	13.0	8.3	6.6	8.7
		MCWBR	9.5	8.9	10.0	9.2	9.9	8.0	7.2	7.7	7.2	5.7	6.3	8.4
	5% WB	MCDBR	9.9	9.4	12.6	13.1	16.3	15.1	13.5	13.5	10.8	7.6	6.4	8.5
		MCWBR	9.5	8.7	10.1	8.4	9.4	7.2	6.7	7.0	6.5	5.7	6.3	8.2
(40) Clear-Sky Solar Irradiance	taub		0.260	0.296	0.318	0.352	0.381	0.421	0.454	0.417	0.343	0.308	0.263	0.238
	taud		1.962	1.998	2.045	2.091	2.263	2.274	2.193	2.267	2.445	2.304	2.014	1.922
	Ebn at Noon		490	680	798	840	845	812	770	771	788	688	482	368
	Edh at Noon		53	88	115	132	120	122	129	111	79	65	49	37
(44) All-Sky Solar Radiation	RadAvg		0.53	1.45	3.01	4.47	5.04	5.65	5.11	4.04	2.44	1.13	0.57	0.30
	RadStd		0.04	0.13	0.25	0.30	0.40	0.58	0.49	0.34	0.24	0.16	0.08	0.03

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page